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Echocardiogram Vector Embeddings Via R3D Transformer for the Improvement of Ejection Fraction Estimation Instruments

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***Abstract*—The estimation of ejection fraction (EF) is critical to intensive care. Doctors commonly use estimated EF to create plans for ICU patients, and a low EF can indicate ventricular systolic dysfunction, which increases the risk of adverse events including heart failure. Recently, interest has grown in deep learning (DL) instruments that can measure cardiac activity to estimate EF automatically. In particular, the vector embeddings learned by DL-based EF estimation models provide valuable mappings of echocardiograms within latent space, which can be utilized to understand the error patterns of DL-based EF estimators and thus improve these instruments. In this work, we provide those embeddings. To this end we repurpose an R3D transformer, a state-of-the-art deep learning model for Video Action Recognition, to classify whether patients have ventricular dysfunction or not (ejection fraction below or above 50%) using echocardiogram data. Our R3D model achieves a test AUC of 0.916 and a test accuracy of 87.5%, approaching the performance of previous comparable studies with a fraction of the training time. Most importantly, the vector embeddings learned by this model will enable future analysis of the errors of DL-based EF estimators, democratizing the improvement of these instruments. The quality of these embeddings, furthermore, is evidenced by the strong results of the model that learned them.**

***Index Terms*—echocardiograms, ejection fraction, vector embeddings**

I. Introduction

Heart failure is the chronic impairment of the heart's ability to effectively pump blood, markedly reducing patients' health and quality of life. Heart failure affects approximately 6.2 million people in the US, and in 2018 alone it was mentioned on over 350,000 death certificates [1]. Moreover, it accounts for $30.7 billion of healthcare expenditures on an annual basis [2]. Fortunately, a range of effective interventions prolong life and reduce morbidity when heart failure exacerbation is detected expediently [3]. Left ventricular dysfunction, caused by damage to or defectiveness of the left ventricle, raises the risk of heart failure [4]. Thus, from a clinical perspective, reliable and expedient diagnosis of ventricular dysfunction is critical to maintaining quality of care.

A key measurement for heart failure diagnoses is ultrasound imagery. In particular, the use of echocardiography, i.e. ultrasound of the heart, allows for the measurement of left ventricular ejection fraction (EF), a key metric in establishing the diagnosis and prognosis of ventricular dysfunction. EF represents the fraction of blood that exits the left ventricle

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during the systolic phase of the cardiac cycle.

Of particular interest are automated echocardiography approaches. Echocardiography is typically interpreted manually by a trained clinician, but interpretation quality is hindered by physician bias and lack of standardization. Additionally, a lack of trained cardiologists in rural or low resource settings can limit this evaluation for patients, leading to downstream negative health effects [2]. Therefore, an instrument for automated EF prediction would significantly facilitate early detection and treatment of ventricular dysfunction, constituting a preventative measure against heart failure [5].

Deep learning can constitute such an instrument, but it entails many challenges. Echocardiograms are stored as video files that suffer issues of high variability, low quality, and lack of standardization. Also burdensome are the demands of videos on memory and storage, as are the limited nature of the two-dimensional imaging modality and the complexity of cardiac physiology and kinesiology [6],[7]. Nevertheless, others have demonstrated the efficacy of different approaches to cardiac pathologies, from video transforms to detect structural heart defects to utilizing CNN-LSTMs to predict cardiomyopathy and cardiac amyloidosis [8],[9].

In this work, we analyzed 10,030 echocardiograms from the EchoNet dataset, released in the EchoNet Dynamic paper, to predict low EF (defined as EF below 50%) utilizing a video transformer. EchoNet represents the largest labeled medical video dataset made available to researchers and medical professionals [10]. Our analysis corroborates the EchoNet Dynamic paper, finding that transformers effectively automate EF determination. Most importantly, we provide the vector embeddings for all 10,030 echocardiograms learned by our model to facilitate the development of similar instruments. This step is necessary to democratize access to echocardiogram representations for clinical ML researchers.

Accurate assessment of ventricular function is critical for patients with heart failure and associated conditions. Therefore, our model and its embeddings serve as a stepping stone towards developing robust clinical decision support instruments that would aid in assessment of patients with suspected heart failure. Potential clinical applications include: (1) automated assessment of ventricular function in low resource settings where trained cardiologists are not available, and (2) expedited, actionable insight in high resource hospital settings where cardiology input is not immediately available.

II. Related Work

Predicting biomedical variables from visual data is, fortunately, tractable. In their I2MTC paper, Mozafari et al. (2022) demonstrated that methods such as tensor decomposition could accurately recover respiratory rate from thermal video data [11]. This shows that tensor-based estimators can extract periodic, biomedical data from tensor representations of videos. In another I2MTC paper, Dandan et al. (2023) demonstrated the ability of U-Net to effectively identify thyroid segments from ultrasound imagery, validating the feasibility of applying deep neural networks to recognize shapes not unlike the left ventricle in ultrasound data [12].

Essential to echocardiogram research in particular was the release of physician-labeled data. The EchoNet Dynamic dataset was published in *Nature* in 2020, as were the three 3D convolutional architectures for video classification used to assess EF as a numerical outcome (ie. percentage). The best of these models, an architecture based on decomposed R2+1D spatiotemporal convolutions, achieved an AUC of 0.97 when utilized for the same binary classification task of predicting whether EF was above or below 50% [10]. Others have also approached the regression problem, predicting continuous EF values, with encouraging results. Previous work by Asch et al. boasted an R-squared of 0.95 on their proprietary database of roughly 50,000 echocardiograms [7].

More recently in 2022, Almadani et al. utilized video action recognition (VAR) neural networks to perform binary classification of a more complex set of echocardiograms. They reported an accuracy of 90.17% with inference time as low as 25.11 seconds using a Gate Shift Network with BNInception as its backbone [13]. Other echocardiogram-based predictive models have also been explored in the literature. Notably, Zhang et al. (2018) were able to detect hypertrophic cardiomyopathy, cardiac amyloidosis, and pulmonary arterial hypertension with C statistics of 0.93, 0.87, and 0.85, respectively, as well as determine left ventricular mass, left ventricular diastolic volume, and left atrial volume within narrow range of machine-derived values [14].

General video transformers have proved useful instruments for measuring other cardiac attributes, too. Jafaeezadeh et al. utilized a deep learning model with inception architecture as the backbone to 71% accuracy in the task of detecting mitral valve dysfunction, evidencing that deep learning on echocardiographic videos can render quick, precise, and stable evaluations of various cardiac pathologies [8]. Dai et. al. applied a novel Cyclical Self-Supervision (CSS) method for learning video-based LV segmentation, which outperformed alternative semi-supervised methods to achieve mean absolute error (MAE) of 4.17, touting state-of-the-art supervised performance [15]. Additionally, CNN-LSTM models have also been applied in this context. Most recently, Hwang et al. achieved an accuracy of 92.8% on detection of LVH (i.e. hypertensive heart disease [HHD], hypertrophic cardiomyopathy [HCM], and light-chain cardiac amyloidosis [ALCA] [9].

III. Methods

A. R3D Transformer Model

The R3D model was among the best-performing approaches that Ouyang et al. (2020) used to tackle the echocardiogram classification problem [10]. Proposed by Tran et al. (2018) as an approach for video learning [16], R3D is especially apt for video data including echocardiograms because it uses 3D convolutional filters. These not only span the 2 spatial dimensions of a frame but also the temporal dimension from stacking them, allowing it to learn patterns not just spatially

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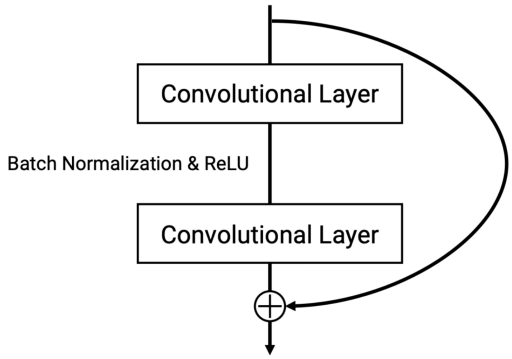


Fig. 1. Example of a shortcut ("skip") connection within a residual block.

within frames but temporally between them. R3D in particular uses spatiotemporal kernels of shape $3 \times 3 \times 3$.

3D CNN architectures are typically shallow, but R3D combines the advantages of spatiotemporal kernels for video learning with those of deeper architectures for better overall performance. In particular, R3D builds on an 18-layer Resnet3D architecture. ResNet owes part of its performance to residual blocks that comprise its architecture: each residual block has a shortcut connection that allows the direct backpropagation of a gradient to earlier layers, which eases gradient calculations and cumulatively mitigates overfitting.

The R3D model we trained originally comes from PyTorch, and it was pre-trained on the Kinetics-400 dataset, which consists of clips of human actions [17]. Our modeling approach thus took advantage of transfer learning by initializing our R3D with weights already tuned to recognize particular actions from human video clips. Although human kinetic actions might seem visually distinct from echocardiograms, similar transfer learning has been successfully utilized in other medical contexts like cancer classification in lung CT images, as was discovered by a 2021 I2MTC paper [18].

In short, R3Ds use spatiotemporal kernels to capture both spatial and temporal patterns in video data, and they rely on the depth of the ResNet18 architecture. When initialized with the weights of pre-existing video transformers, R3D performance can also benefit from the "head start" of transfer learning. They are therefore powerful tools for video classification.

B. Training

Training was prohibitively expensive on local devices due to memory constraints. Initial mitigation strategies included downscaling all videos to a resolution of 56×56 instead of 112×112 and grayscaling all videos from 3 channels (RGB) to 1 (gray). Our final models, however, capitalized on cloud computing resources and were able to utilize the training data without downscaling or grayscaling. We also downsampled the videos by only using 1 of every 4 frames when training and testing our final model. Both the MIT Satori cluster and Google Cloud Platform (GCP) provided computational resources for model training. Model performance peaked at 4 epochs of training, declining from overfitting in subsequent epochs. Transfer learning likely contributed to such speed of convergence on the part of the model.

We used an Adam optimizer and achieved our best results based on validation sets using a learning rate of 1e-5 and a weight decay of 5e-4. To further manage our memory resources, we also used gradient accumulation, which involved accumulating gradients over successive mini-batches of size 20 before using them to update the model. This comes with computational costs, as there are more forward and backward passes involved with each parameter update. Nevertheless, memory was a tighter constraint, which motivated the final adoption of gradient accumulation in our modeling process. We also chose gradient accumulation as opposed to batching because the echocardiogram videos varied in frame count.

C. Classification Metrics

This problem is a binary classification one. As such, binary cross entropy loss was used to train the classification model. Accuracy, the ratio of true predictions to all predictions, was a necessary metric in order to compare the results of our model with those from related literature. It can become misleading, however, when the label distribution is imbalanced, which was the case for ventricular dysfunction.

This motivated the use of more nuanced metrics. Precision (TP / (TP + FP)) describes the ability of a classifier to identify only relevant (positive) data points. Recall (TP / (TP + FN)) describes the ability of a classifier to identify any relevant data points to begin with. F1 score is the harmonic mean of both. Specificity (TN / (TN + FP)) conversely measures how well a classifier identifies negative data points. Given the importance of detecting ventricular dysfunction (positive class) early, we aimed primarily to maximize recall.

In addition to AUC we recorded AUPRC, or the area under the precision-recall curve. This is because a small number of correct or incorrect predictions can result in large changes in the ROC curve for imbalanced data. This is not the case for the precision-recall curve, making AUPRC a better holistic measure of discriminative ability for the type of imbalanced dataset at hand.

IV. Cohort

A. Cohort Selection

Our dataset consists of 10,030 echocardiograms. Ejection fraction (EF) served as our target variable, and we were interested in whether it was above or below 50%. This is because an EF below 50% warrants certain medical treatments that an EF above 50% does not. This caused a class imbalance program, as nearly 78% of the EchoNet echocardiograms had an EF that was greater than or equal to the 50% cutoff. Only 22% were below. The precise figures can be seen in Table 1.

The distribution of EF in our dataset was left-skewed with a mode around 60%. Specifically, the mean EF is 0.55, which is slightly lower, because there is a group of EFs between 20-50% that influence the average. Notably, there were virtually

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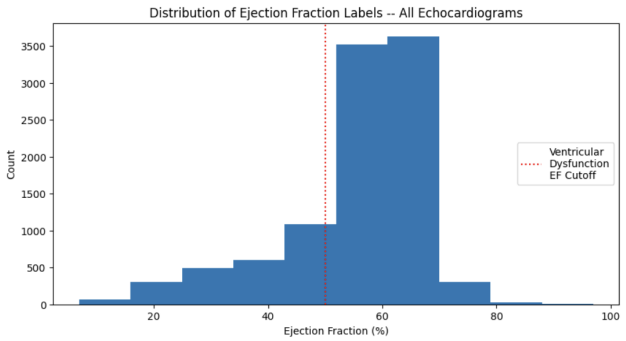


Fig. 2. Distribution of Ejection Fraction for All Echocardiograms

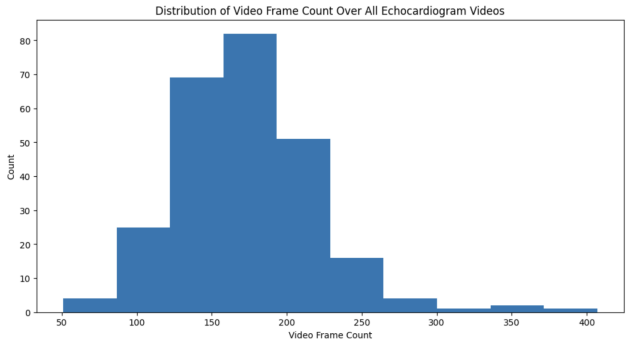


Fig. 3. Distribution of Frame Length for All Echocardiograms

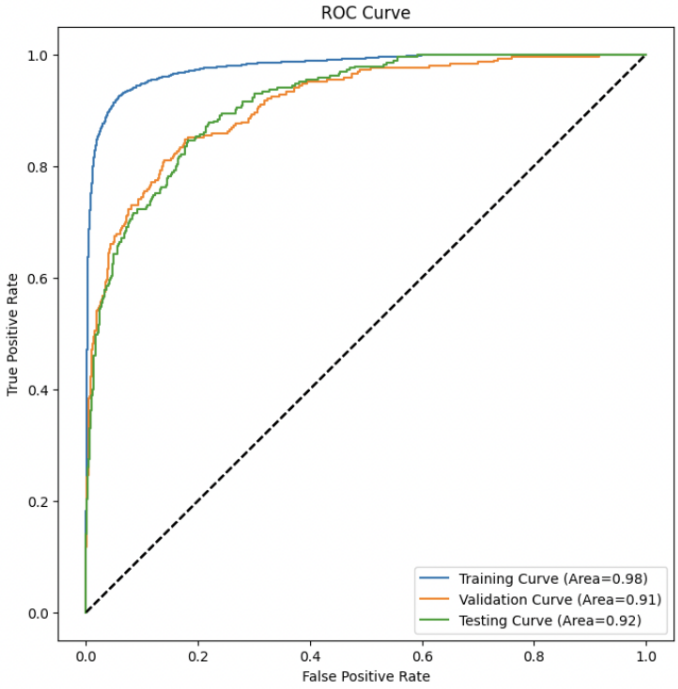


Fig. 4. AUC-ROC Curve for Train, Val, and Test Sets

no records where EF exceeded 80% or fell below 10%. The standard deviation is 12.

TABLE I
COHORT SPLIT BASED ON EJECTION FRACTION (EF)

Class	Description	Num Videos	Name
1	EF<50%	2,246	Unhealthy
0	EF≥50%	7,784	Healthy

Within the EchoNet dataset, echocardiograms have 173 frames on average with a standard deviation of 47 frames. This distribution appears normal. We chose not to standardize frame lengths to preserve variations in frame length for all videos in training, validation, and test sets. Keeping a variable frame length was important, as it reflects the medical reality that echocardiograms are measured in different lengths. Training a deep learning model on echocardiograms of frame length 170, for example, may not generalize well to echocardiograms of frame length 50 or 400. Non-standardized frame lengths thus functioned as a robustness feature.

Our dataset split resulted in 7,458 training videos, 1,284 validation videos, and 1,277 test videos. There was an underlying class imbalance wherein only 1,672 (22.4%) of the training videos were class 1 (EF<50%).

V. RESULTS

The performance of our models is on par with most state-of-the-art models, falling just 5 points short of the 0.97 AUC of the leading EchoNet Dynamic model [10]. This discrepancy arises from several sources. One is that each training echocardiogram is downsampled by a factor of 4, such that the processed training video contains one of every 4 frames from the original. This was done to work around limited computational resources, but it also reduced the volume of the data that our R3D model was trained on, thus weakening performance. Furthermore, we did not augment the training data of our model as was done in the EchoNet Dynamic model, again for computational reasons. Data augmentations such as brightening and jittering the training videos have been shown to strengthen the performance of echocardiogram-based classifiers, as they increase the volume of and variation within the training data [13].

TABLE II
R3D TRANSFORMER MODEL RESULTS

Model Metric	Data Split		
	Training	Validation	Testing
Accuracy	0.951	0.889	0.875
AUC	0.979	0.912	0.916
Precision	0.924	0.809	0.789
Recall	0.85	0.661	0.604
Specificity	0.98	0.955	0.954
F1 Score	0.885	0.728	0.684
AUPRC	0.949	0.811	0.798
BCE Loss	0.151	0.306	0.31

The AUPRC of our R3D was also strong. The AUPRC of a random classifier is theoretically 0.5, and for a perfect classifier it should be 1.0. The R3D test score was 0.798. Given how AUC can be misleading in contexts of class imbalance,

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such a high AUPRC value suggests it is genuinely the strength of the R3D classifier itself that explains the high AUC.

TABLE III
COMPARISON OF TEST ACCURACY AND AUC WITH EARLIER CLASSIFIER BY ALMADANI ET AL. (2022)

Model	Classification Approach	
Metric	*Our Approach*[a]	*Almadani et al.*[b]
Accuracy	87.5%	90.17%
AUC	0.916	0.847
EF Classification Threshold	50%	50%

[a]R3D transformer with ResNet18 architecture, transfer learning
[b]GSM with inception backbone, 32-frame echocardiograms

Test specificity was especially strong at 0.954. This means the classifier makes negative predictions correctly almost every time. Test recall was lower at 0.604, but this still means that the model predicts the majority of true positives (cases of ventricular dysfunction) correctly. Test precision, finally, was also high at 0.789. This means that a comfortable majority of positive predictions made by the R3D model were in fact true cases of ventricular dysfunction. Together, these metrics demonstrate the strong discriminative ability of this classifier to predict EF being above or below 50%. The vector embeddings of this model, therefore, are reflective of an effective learned representation of echocardiogram data.

VI. DISCUSSION

A. Comparison to State-of-the-Art Echocardiography Models

In 2022, Almadani et al. reported their utilization of video action recognition (VAR) neural networks to perform binary classification of echocardiograms [13]. A comparison of the AUC and accuracy of our R3D Transformer to that of the VAR model used by Almadani et al. is visible in Table III.

"Deep Video Action Recognition Models for Assessing Cardiac Function from Echocardiograms" achieved its highest accuracy of 90.17% with 32 frames on the GSM with BNInception. Its highest AUC of 0.847 was achieved with 16 frames on the GSM with InceptionV3. Our R3D model achieved a comparable accuracy of 87.5% and an even higher AUC of 0.916, demonstrating that it has at least the holistic discriminative ability of state-of-the-art ventricular dysfunction classifiers and has competitive accuracy. This is important to demonstrate, as it means that the vector embeddings learned by our model are of a quality consistent with the best classifiers in the current literature.

B. Additions for future development

Gradient weighted class activation mapping (Grad-CAM) use gradients to visualize regions of an image that most influence neural network predictions, displaying them as a heatmap. Future work will use Grad-CAM to conduct error analysis and investigate whether incorrectly classified records are clustered together in embedding space.

C. Applications of Vector Embeddings to Instrumentation

The vector embeddings obtained from our R3D model enable diverse applications to cardiac care instruments. This is because they encapsulate rich information about echocardiograms in a latent space, enabling various downstream tasks and analyses. We describe a few of these applications here.

Echocardiogram-based Clustering: Clustering analyses using echocardiogram vector embeddings could reveal patterns and subtypes within cardiac data that represent diverse presentations of ventricular dysfunction. Active sampling of echocardiograms from a diversity of these clusters could robustify the training of current EF estimation instruments.

Disease Classification Instruments: Vector embeddings can train classification instruments to automatically identify specific cardiac conditions beyond EF. This is because a lower EF can be caused by other conditions like coronary artery disease or systolic heart failure [19]. This extends the model's measurement utility to a broader spectrum of cardiac pathologies.

Patient Risk Measurement: By incorporating vector embeddings, new risk prediction models could assist in stratifying patients based on their risk of developing ventricular dysfunction or related complications, enabling targeted interventions and personalized treatment plans.

New Cardiac Measurements for Clinical Decision Support Systems: Vector embeddings can provide healthcare professionals with additional contextual information for interpreting echocardiograms, constituting a new form of cardiac measurement. The collaborative approach in integrating these into either models and systems combines the strengths of deep learning with the expertise of clinicians.

D. Clinical Implications

On multiple metrics, the R3D transformer-based approach used in this study effectively predicts low EF. This means it can enhance shared decision-making which is a crucial element in patient-centric care, and it demonstrates that deep learning approaches can prove successful as an instrument for the clinical prediction of ventricular dysfunction.

To note, clinical decision making for a ventricularly dysfunctional patient is complicated, requiring contextualization of EF. For example, implantable cardioverter-defibrillators (ICDs) are considered for primary prevention in patients at high risk for sudden cardiac death; an EF$\leq$35% is a clinically accepted predictor for elevated risk and strong benefit from an ICD implant. Such treatments may be considered in patients with clinical heart failure with an EF$\leq$35% for primary prevention of sudden death. Hence identification of ICD candidates is important in improving patient outcomes. For patients who might be at increased risk of ICD, insertion is detrimental for patient outcomes. In addition to predicting EF$<$50% relative to other counterparts, our model performance may generalize to predict EF$\leq$35% which could facilitate treatment selection in these patients. Likewise, the architecture used here can be extrapolated to predict improvement of EF following medical therapy to decide more rapidly on primary device placement in those unlikely to show an EF increase to $\leq$35%.

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Deep learning models are best utilized as decision support tools. In particular, contextualization of EF predictions by a cardiologist is crucial for comprehensive clinical decision-making in a heart failure patient. An EF prediction alone may not capture the entire clinical reality. Additional factors such as the patient's symptoms, prior response to medical therapy, and individualized treatment goals must be considered in real world clinical practice. Deep learning models can be integrated with the clinical expertise of cardiologists by considering both their predictions and the patient's unique circumstances. Therefore these models have the potential to guide a more targeted approach to treatment selection addressing care gaps for high-risk individuals, and potentially reducing the need for costly and unnecessary procedures.

VII. Conclusion

In this paper, we provide the vector embeddings of an R3D deep video action recognition network that we trained to classify the healthiness of a heart based on estimated EF from ultrasound measurements. This model achieves a high AUC and accuracy, on par with the performance of current best-practice models, and the strength of these results serves to legitimize the quality of the vector embeddings published in this work. At present, these vector embeddings describe each of the 10,030 echocardiograms in the EchoNet dataset in 400-dimensional latent space. These results constitute valuable lookup points for future error analyses on automatic EF estimators and thus for new improvements to these instruments. Our hope is that future virtual instruments that use ultrasound measurements of the heart can better estimate EF because their development utilized the vector embeddings we publish here.

VIII. Code

The vector embeddings found by this paper can be accessed through the following GitHub repository: https://github.com/Team-Echo-MIT/r3d-v0-embeddings

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